Magnetospheric formation processes of diffuse aurora

Report

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1 Introduction

Visible emissions called aurorae are caused when energetic electrons collide with atoms in the atmosphere, leading to the ionization and excitation of the atmospheric constituents. In general, the phenomenon of planetary aurorae consists of two components, the discrete and the diffuse aurora, which differ not only in their optical appearance but also in the underlying physical mechanisms leading to their generation. The discrete aurora, referring to localized regions of intense auroral arcs, is associated with field-aligned currents leading to the acceleration of the current-carrying particles. Diffuse aurora, in contrast, is a broader region of glow without large-scale structures and produced through the unmodified precipitation of magnetospheric plasma. More specifically, it is known that resonant wave-particle interactions lead to the scattering of plasma sheet electrons with energies $E \sim 0.1\,\mathrm{keV}$ to $10\,\mathrm{keV}$ which in turn cause the optical phenomenon of diffuse aurora.
The Earth‘s diffuse auroral precipitation provides the main source of energy input into the high-latitude nightside upper atmosphere. Thus, a better understanding of diffuse aurora is relevant because, as a space weather phenomenon, it affects both technical infrastructure and potentially our climate.

2 Theory

The diffuse aurora extends around the entire auroral oval, which lies at high latitudes around 65° to 70° north or south of the equator. In general, it can be seen as a belt of weak emissions extending over a latitudinal range of 5° to 10°, which intensifies with increasing geomagnetic activity level (Petrinec et al., 1999). Furthermore, there are variations in intensity with solar wind conditions as well as seasonal changes. The highest photon flux can be measured in the magnetic local time (MLT) sector from pre-midnight to dawn, which is related to the combination of eastward transport of electrons due to the $E \times B$ drift and gradient drifting from the nightside plasma sheet. The diffuse aurora is primarily caused when low-energy electrons in the central plasma sheet are scattered into the atmospheric loss cone and then subsequently removed at low altitude by collisions with the atmospheric constituents. The region of occurrence therefore maps along the magnetic field lines from the outer radiation belts ($L \sim 4$) to the entire central plasma sheet ($L \sim 12$).

2.1 Adiabatic Invariants

In general, particles which are trapped in the geomagnetic field undergo three types of cyclic motion. These are cyclotron motion about the magnetic field, bounce motion between the mirror points in the northern and southern hemisphere, and azimuthal drift around the Earth. With each quasi-periodic motion, an approximate conservation law is associated, called the adiabatic invariants, which are conserved if the system changes slowly. If there are temporal variations comparable to the drift, bounce or cyclotron frequency, the associated adiabatic invariants are violated.

The first adiabatic invariant, often called the magnetic moment, is associated with the cyclotron motion around the magnetic field. It is given by

$$\mu = \frac{p^2 \sin^2 \alpha}{2 m_0 B}, \tag{1}$$

where p is the momentum of a particle, α is the pitch angle, m_0 is the particle's rest mass and B is the background magnetic field. The timescale of cyclotron motion is less than a millisecond for electrons, corresponding to a frequency of a few kHz. As the particle moves along the field line, its pitch angle changes and the conservation of μ allows us to relate the pitch angle at higher latitudes to that at the equator. If the pitch angle of the particle is smaller than the loss cone angle α_{lc}, the particle is lost to the atmosphere, otherwise (if $\alpha > \alpha_{\mathrm{lc}}$) it will remain trapped.
The second adiabatic invariant, associated with the bounce motion between the northern and southern hemisphere, is given by

$$J = \int_{bounce} p_{\parallel} \, \mathrm{d}s, \tag{2}$$

where $p_{\parallel}$ refers to momentum parallel to the magnetic field and s is the distance along the particle's bounce trajectory. The timescale of the bounce motion is typically a few tenths of a second for MeV electrons.
The third adiabatic invariant is associated with the drift period around the Earth and given by

$$\Phi = \oint_{drift} B \, \mathrm{d}S, \tag{3}$$

where S is the azimuthal drift path, meaning that the magnetic flux enclosed by the drift orbit is conserved. For MeV electrons, the timescale of the drift motion is typically around $10 - 15\,\mathrm{min}$.
If the Earth's magnetic field changes very slowly, all three adiabatic invariants are conserved and there is no net acceleration or loss of the particles. If the magnetic field is changing at frequencies comparable to the drift, bounce or cyclotron frequencies, the corresponding invariants are broken and there will be loss or acceleration of particles. As the timescales of the adiabatic invariants are widely separated, field fluctuations on e.g. a timescale of minutes can break the third invariant, but still conserve the first and second invariants.
Particle precipitation into the atmosphere is generally associated with the violation of the first adiabatic invariant, such that the pitch angle of initially trapped particles approaches the loss cone angle and the particles are lost to the atmosphere.

2.2 Wave-particle interactions

In the magnetosphere, many types of plasma waves can occur. They can be generated when the electron distribution becomes unstable leading to instabilities in the plasma. The precipitation of electrons causing the diffuse aurora is primarily attributed to wave-particle interactions, with the particles' pitch angles being altered by electromagnetic

fields changing the first adiabatic invariant. In general, waves can resonate with particles when the Doppler-shifted wave frequency experienced by the particle equals a multiple of the electron gyrofrequency,

$$\omega - kv\cos\theta\cos\alpha = \frac{n\Omega_\sigma}{\gamma} \qquad n = 0, \pm1, \pm2, \ldots, \tag{4}$$

where ω is the wave frequency, k is the wave number, θ is the wave normal angle with respect to the ambient magnetic field B_0, v is the particle velocity, α is the pitch angle of the particle and $\Omega_\sigma = qB_0/m_\sigma$ is the non-relativistic particle gyrofrequency for the particle species σ of charge q and rest mass m_σ, and γ is the Lorentz factor.
In Equation 4, $n = 0$ corresponds to the Landau resonance $\omega = k_\parallel v_\parallel = kv\cos\theta\cos\alpha$, and $n \neq 0$ gives the gyroresonances. The resonant velocity in the non-relativistic case ($\gamma = 1$) is

$$v_{\parallel,res} = (\omega - n\Omega_\sigma)/k_\parallel. \tag{5}$$

This resonance condition is independent of the perpendicular velocity $v_\perp$ of the particle and defines a straight line in the $(v_\parallel, v_\perp)$-plane. For the relativistic case, the resonance condition defines an ellipse, e.g. for $n = 1$:

$$v_{\parallel,res} = -\frac{\omega}{k_\parallel}\left(1 - \frac{\Omega_\sigma}{\omega}\frac{1}{c}\sqrt{c^2 - v_{\parallel,res}^2 - v_{\perp,res}^2}\right). \tag{6}$$

In the case of Landau resonance, $n = 0$, the resonance condition is always $v_{\parallel,res} = \omega/k_\parallel$, defining a straight line also in the relativistic case. In order to solve the resonance condition and find the velocities and energies of the particles that the wave will interact with, we need to know the cyclotron frequency Ω_σ, wave frequency ω, wave propagation angle θ and we must solve the plasma dispersion relation relating the wave frequency ω to the wave vector k. The resonant ellipse then shows the energy and pitch angle of the electrons that a wave of a specific frequency will interact with.
In the rest frame of the particles, they move along surfaces defined by

$$v_\perp^2 + (v_\parallel - \omega/k_\parallel)^2 = \text{const.}, \tag{7}$$

which are circles centered at $v_\parallel = \omega/k_\parallel$ (Koskinen and Kilpua, 2022). These circles (or ellipses in the relativistic case) are known as single wave characteristics. Resonance occurs when the single-wave characteristics cross the resonant ellipse. At the resonant velocity $v_{\parallel,res}$, the particles move along the single wave characteristics towards regions of lower phase space density. Therefore, wave and particle can efficiently exchange energy and momentum, which potentially results in wave amplification or damping. Whether the wave grows or is damped depends on the behavior of the particle distribution function near the resonant velocity. If the particles move to smaller pitch angles and lose energy, the wave gains energy. In order to determine the net wave growth or decay, the full dispersion relation needs to be solved.

2.3 Particle Scattering and Diffusion

The combined effect of several resonant interactions between the wave and particles is called diffusion of the particle distribution in the velocity space and can lead to damping or growth of the wave. If there is a broad band of waves, the resonances overlap and the process can be treated as a diffusion problem. In the quasilinear diffusion approach, we assume that there is a small pitch angle scattering by each wave and that the diffusion is proportional to the wave power. The waves are assumed to be uncorrelated and there are no trapping effects.
The temporal evolution of the phase space density f can be described by the Fokker-Planck equation (Subbotin and Shprits, 2012),

$$\frac{\partial f}{\partial t} = \sum_{i,j} \frac{\partial}{\partial J_i} D_{J_i J_j} \frac{\partial f}{\partial J_j}, \tag{8}$$

where $J_1 = \mu$, $J_2 = J$ and $J_3 = \Phi$ are the adiabatic invariants described in Section 2.1. $D_{J_i J_j}$ are the diffusion coefficients due to wave-particle resonant interactions which are typically calculated in terms of (p, α, L). Lyons et al. (1972) give the general form for bounce-averaged quasi-linear diffusion coefficients $\langle D_{\alpha\alpha} \rangle$ (pitch-angle diffusion), $\langle D_{\alpha p} \rangle$ (mixed diffusion) and $\langle D_{pp} \rangle$ (momentum diffusion) as

$$\langle D_{\alpha\alpha} \rangle = \frac{1}{\tau_B} \int_0^{\tau_B} D_{\alpha\alpha}(\alpha) \left(\frac{\partial \alpha_{eq}}{\partial \alpha} \right)^2 \, \mathrm{d}t, \tag{9}$$

$$\langle D_{\alpha p} \rangle = \frac{1}{\tau_B} \int_0^{\tau_B} D_{\alpha p}(\alpha) \left(\frac{\partial \alpha_{eq}}{\partial \alpha} \right) \, \mathrm{d}t, \tag{10}$$

$$\langle D_{pp} \rangle = \frac{1}{\tau_B} \int_0^{\tau_B} D_{pp}(\alpha) \, \mathrm{d}t. \tag{11}$$

Here, $D_{\alpha\alpha}$, $D_{\alpha p}$ and D_{pp} are local diffusion coefficients, α and α_{eq} are local and equatorial pitch angle, respectively, and τ_B is the electron bounce period.

3 Whistler-mode chorus waves

Previous theoretical studies (Thorne et al., 2010; Ni et al., 2008) have indicated that there are two distinct classes of plasma waves, whistler-mode chorus waves and electrostatic Electron Cyclotron Harmonic waves, which can be responsible for the underlying pitch angle scattering of electrons. It has been shown that both the spectral intensity of the chorus waves and the ECH waves have a geomagnetic activity variation and an MLT distribution similar to the diffuse aurora (Thorne et al., 2010).
Whistler-mode chorus waves are excited in the region outside the plasmasphere during the injection of plasma sheet electrons. They occur characteristically in two separate frequency bands with a gap at half of the electron gyrofrequency, which are often referred to as lower band and upper band chorus, respectively. Recent theoretical and numerical studies (Thorne et al., 2010; Ni et al., 2011a) have demonstrated that whistler-mode chorus waves are the dominant contributor to the formation of nightside diffuse aurora

in the inner magnetosphere (L < 8). The combination of upper band and lower band chorus causes rapid pitch angle and momentum scattering of electrons at energies of a few keV over a broad range of pitch angles and can reproduce the observed remnant pancake electron distribution (Thorne et al., 2010).
Quasi-linear bounce-averaged diffusion coefficients as given in Equations 9, 10 and 11 can be calculated for whistler-mode chorus waves using the Full Diffusion Code (Ni et al., 2008; Shprits and Ni, 2009; Orlova and Shprits, 2011). The workflow of the code is sketched in Figure 1. In this code, the quasi-linear bounce-averaged diffusion coefficients are calculated using a dipole magnetic field in the form given by Glauert and Horne (2005):

$$\langle D_{\alpha\alpha}\rangle = \frac{1}{S(\alpha_{eq})}\int_0^{\lambda_m} D_{\alpha\alpha}(\alpha)\frac{\cos\alpha}{\cos^2\alpha_{eq}}\cos^7\lambda\ \mathrm{d}\lambda, \tag{12}$$

$$\langle D_{\alpha p}\rangle = \frac{1}{S(\alpha_{eq})}\int_0^{\lambda_m} D_{\alpha p}(\alpha)\frac{\cos^4\lambda\left(1+3\sin^2\lambda\right)^{1/4}}{\cos\alpha}\ \mathrm{d}\lambda, \tag{13}$$

$$\langle D_{pp}\rangle = \frac{1}{S(\alpha_{eq})}\int_0^{\lambda_m} D_{pp}(\alpha)\frac{\cos\lambda\left(1+3\sin^3\lambda\right)^{1/2}}{\cos\alpha}\ \mathrm{d}\lambda, \tag{14}$$

where λ_m is the mirror latitude and $S(\alpha_{eq})$ gives the variation of τ_B with α_{eq}, approximated by $S(\alpha_{eq}) = 1.3 - 0.56\sin\alpha_{eq}$ in a dipole magnetic field (Hamlin et al., 1961). α_{eq} is associated with the local pitch angle by $\sin^2\alpha = \frac{\sqrt{4-3\cos^2\lambda}}{\cos^6\lambda}\sin^2\alpha_{eq}$.
λ_m is the upper limit of the magnetic latitude determined either by the mirror latitude of the particles, which is found by solving the polynomial

$$C_l^6 + 3C_l\sin^4\alpha_{eq} - 4\sin^4\alpha_{eq} = 0 \tag{15}$$

for $C_l = \cos^2\lambda_m$, or by the maximum latitude of the wave occurrence.

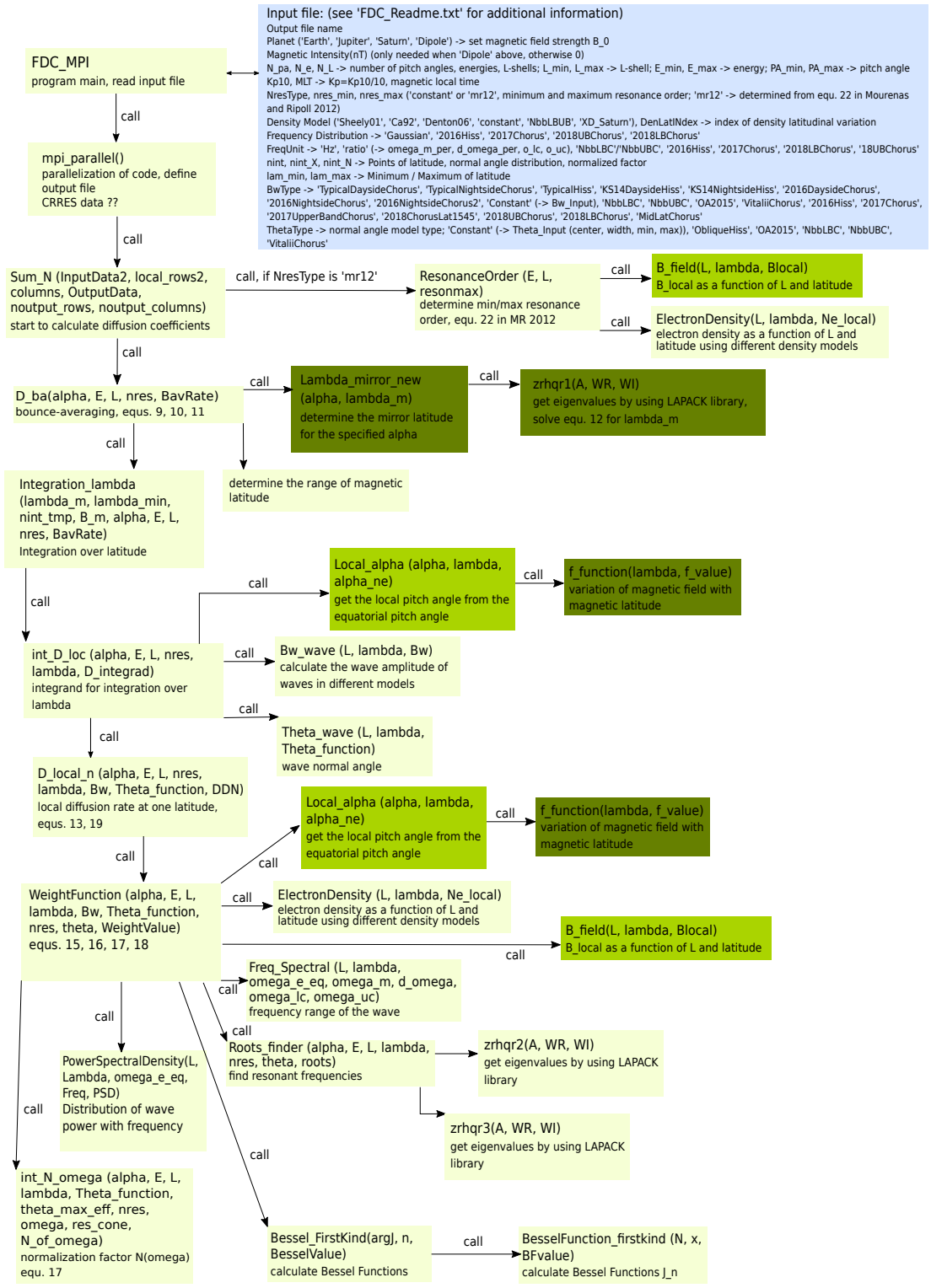


Figure 1: Full Diffusion Code for the calculation of quasi-linear bounce-averaged diffusion coefficients for wave-particle interaction with whistler-mode chorus waves.

The local pitch angle diffusion rate $D_{\alpha\alpha}$ is calculated following Albert (2007):

$$D_{\alpha\alpha} = \frac{\Omega_\sigma}{\gamma^2}\frac{B^2_{\rm wave}}{B_0^2}\sum_{n=-\infty}^{\infty}\sum_{\omega} D^n_{\alpha\alpha}, \tag{16}$$

$$D^n_{\alpha\alpha} = \int_{\theta_{\rm min}}^{\theta_{\rm max}} \sin\theta \ {\rm d}\theta \Delta_N G_1 G_2, \tag{17}$$

$$\Delta_N(\omega,\theta) = \frac{\pi}{2}\frac{\sec\theta}{|v_\parallel/c|^3}\Phi_n^2\frac{\left(-\sin^2\alpha + n\Omega_\sigma/\omega\gamma\right)^2}{|1-\left(\partial\omega/\partial k_\parallel\right)_\theta/v_\parallel|}, \tag{18}$$

$$G_1(\omega) = \frac{|\Omega_\sigma|B^2(\omega)}{\int_{\omega_{LC}}^{\omega_{UC}} B^2(\omega')\ {\rm d}\omega'}, \qquad G_2(\omega,\theta) = \frac{g_\omega(\theta)}{N(\omega)}, \tag{19}$$

$$N(\omega) = \int_{\theta_{\rm min}}^{\theta_{\rm max}} {\rm d}\theta' \sin\theta' \Gamma g_\omega(\theta'), \qquad \Gamma = \mu^2\left|\mu + \omega\frac{\partial\mu}{\partial\omega}\right|, \tag{20}$$

$$\begin{aligned}\Phi_n^2 = &\left[\left(\frac{D}{\mu^2-S}\right)^2\left(\frac{\mu^2\sin^2\theta - P}{\mu^2}\right)^2 + \left(\frac{P\cos^2\theta}{\mu^2}\right)^2\right]^{-1} \\ &\cdot\left[\frac{\mu^2\sin^2\theta - P}{2\mu^2}\left(1+\frac{D}{\mu^2-S}\right)J_{n+1} + \frac{\mu^2\sin^2\theta - P}{2\mu^2}\left(1-\frac{D}{\mu^2-S}\right)J_{n-1}\right. \\ &\left. + \cot\alpha\sin\theta\cos\theta J_n\right]^2\end{aligned} \tag{21}$$

Here, $\mu = kc/\omega$ is the refractive index given as a function of (ω,θ) by cold plasma theory (Stix, 1962). The term Φ_n^2 accounts for the relationships between the components of the wave electric and magnetic fields as well as the gyroaveraging of the resonant wave-particle phase. It contains Bessel functions J_n and $J_{n\pm1}$ with arguments $k_\perp p_\perp/m_\sigma\Omega_\sigma$ and the Stix coefficients D, S, P.

The function $B^2(\omega)$ describes the frequency distribution of wave power and is taken to be zero unless ω lies between the lower and upper cutoffs ω_{LC} and ω_{UC}. It can be specified in the input file and in the subroutine `Bw_wave.f90`.

Similarly, the distribution of wave power with normal angle θ is described by $g_\omega(\theta)$, which is zero unless θ lies between $\theta_{\rm min}$ and $\theta_{\rm max}$. It can be specified in the input file and in the subroutine `Theta_wave.f90`.

$G_2(\omega,\theta)$ and $\Delta_N(\omega,\theta)$ are computed at the resonant frequency ω corresponding to θ and n, determined from the cyclotron resonance condition, Equation 4. There may be several such values of ω as simultaneous solutions of Equation 4 and the cold plasma wave dispersion relation.

Local cross diffusion rate $D_{\alpha p}$ and momentum diffusion rate D_{pp} can be subsequently obtained by

$$D_{\alpha p} = D_{\alpha\alpha}\left[\frac{\sin\alpha\cos\alpha}{n\Omega_\sigma/(\gamma\omega_k) - \sin^2\alpha}\right] \qquad D_{pp} = D_{\alpha\alpha}\left[\frac{\sin\alpha\cos\alpha}{n\Omega_\sigma/(\gamma\omega_k) - \sin^2\alpha}\right]^2. \tag{22}$$

4 Electron cyclotron harmonic waves

ECH waves are electrostatic emissions observed in narrow frequency bands between the harmonics of the electron gyrofrequency. They are thought to be excited by the loss cone instability of the ambient plasma sheet electrons and intensify with increasing geomagnetic activity (Ni et al., 2011b). While the occurrence rate of moderately intense chorus emissions drops significantly in the outer magnetosphere ($L > 8$), moderately strong ECH emissions can still be observed up to $L \sim 12$ (Zhang et al., 2013). This suggests that scattering by ECH waves could be an important and even major driver of the nightside electron diffuse auroral precipitation in the outer magnetosphere (Zhang et al., 2015). Furthermore, ECH waves are possibly connected to the occurrence of pulsating aurora especially at high altitude (Liang et al., 2010).

4.1 Latitudinal distribution of ECH Waves

The determination of ECH wave-induced bounce-averaged diffusion coefficients requires information about the wave propagation characteristics, such as wave normal angle and wave power spectrum along the particle bounce trajectory. Ni et al. (2011b) use the ray tracing code HOTRAY to trace the waves in a hot, magnetized, linearly unstable plasma. The code requires the specification of the spatial distribution of the background magnetic field and plasma density, and properties of the hot plasma sheet electrons responsible for the wave excitation, such as temperature and loss cone parameters. Ni et al. (2011b) assume a dipole magnetic field model and use CRRES measurements near $L = 6$ for equatorial values of the ambient magnetic field and the cold plasma density under different levels of geomagnetic activity. They model the electron distribution by a sum of subtracted bi-Maxwellian components given by

$$\begin{aligned} f_i(v_\perp, v_\parallel) = \frac{n_i}{\pi^{3/2}\alpha_{\perp,i}^2\alpha_{\parallel,i}} \exp\left(-\frac{v_\parallel^2}{\alpha_{\parallel,i}^2}\right) \cdot \left[\Delta_i \exp\left(-\frac{v_\perp^2}{\alpha_{\perp,i}^2}\right)\right. \\ \left. + \frac{1-\Delta_i}{1-\beta_i}\left(\exp\left(-\frac{v_\perp^2}{\alpha_{\perp,i}^2}\right) - \exp\left(-\frac{v_\perp^2}{\beta_i\alpha_{\perp,i}^2}\right)\right)\right], \end{aligned} \tag{23}$$

where n_i is the electron density, $\alpha_{\perp,i}$ and $\alpha_{\parallel,i}$ are the thermal velocities perpendicular and parallel to the ambient magnetic field and Δ_i and β_i are the loss cone parameters. Ni et al. (2011b) investigate an ECH excitation event during geomagnetically active conditions and use five components of the form 23 to fit the electron pitch angle distribution over velocity space observed by CRRES. From this fit, they obtain the parameters for the hot electron components listed in Table 1.

Table 1: Hot electron components used by Ni et al. (2011b) to model the electron distribution function during geomagnetically active and moderate times at $L = 6$.

Component	$T_\perp$ / eV	$T_\parallel$ / eV	N_e / m^{-3}	Δ	β
1	151	91	1.6×10^4	0.9	0.9
2	721	680	4.0×10^4	0.6	0.5
3	1821	1646	2.0×10^5	0.3	0.1
4	9319	5126	3.8×10^5	0.3	0.9
5	12609	10366	1.2×10^5	0.9	0.4

For the cold plasma component, they assume $T_\perp = T_\parallel = 1\,\text{eV}$ and loss cone parameters $\Delta = 1$ and $\beta = 0.5$. The cold plasma density is obtained from CRRES measurements. For quiet geomagnetic activity conditions, they adopt the parameters obtained by Horne et al. (2003).
Ni et al. (2011b) use the parameters obtained from modeling the electron distributions as input to the HOTRAY code to solve the hot plasma dispersion relation for four specific frequencies in each harmonic band by varying the wave normal angle θ (WNA) at the magnetic equator. They conclude that the temporal growth rates change significantly with WNA and that the growth rates peak at WNAs closer to $90\,^\circ$ for higher harmonic bands. Within each harmonic band, they find that the WNA corresponding to the peak growth rate is smallest near the central frequency and migrates nearer to $90\,^\circ$ with decreasing and increasing frequency. Because the peak growth rate occurs at different WNAs for different frequencies, they expect that the waves at each frequency reach different maximum magnetic latitudes. Based on their results from HOTRAY calculations of the variation of the latitude along the raypath, Ni et al. (2011b) establish the following model for the WNA distribution:

- The wave is assumed to be confined within $3\,^\circ$ of the magnetic equator. The equatorial WNA of each wave frequency is defined by the WNA of the peak wave growth rate θ_e, which in turn defines the concrete latitudinal range for each wave frequency.

- Specifically, the maximum latitude up to which the wave is assumed to propagate is given by
  $\lambda_{\max} = 3\,^\circ$ if $\theta_e < 89.5\,^\circ$
  $\lambda_{\max} = 2\,^\circ$ if $89.5\,^\circ \leq \theta_e < 89.8\,^\circ$
  $\lambda_{\max} = 1\,^\circ$ if $\theta_e \geq 89.8\,^\circ$
  Linear interpolation is applied between θ_e at the magnetic equator and $\theta = 90\,^\circ$ at the boundary latitude $\lambda_{\max}$ to construct the latitudinal variation of the WNA during propagation.

- The WNA model is defined by the WNA θ, the WNA width $\delta\theta = 90\,^\circ - \theta$, the perpendicular and parallel components of the wave number $k_{0,\perp}$ and $k_{0,\parallel}$ and the width of the parallel wave number $\delta k_\parallel$. $k_{0,\parallel}$ is obtained by solving the hot plasma dispersion relation at the equator with HOTRAY, while the perpendicular wave

number $k_{0,\perp}$ is assumed to be constant.
$\delta k_\parallel$ is then given by $\delta k_\parallel = k_{0,\perp}/\tan(\theta - \delta\theta) - k_{0,\parallel}$.

These wave parameters, λ, θ, $k_\perp$, $k_\parallel$, $\delta\theta$ and $\delta k_\parallel$ provide the input for the calculation of the bounce-averaged diffusion coefficients with a modified version of the Full Diffusion Code.
Similar to the method described above, Lou et al. (2022) use the WHAMP (Waves in homogeneous, anisotropic, multicomponent plasmas) Code to solve the hot plasma dispersion relation and calculate wave number and temporal growth rates as a function of wave frequency. They assume WNAs between 86 ° and 89.96 ° and use the model for the WNA distribution described above. By combining the results from WHAMP with the WNA model, the latitudinal variation of the wave parameters θ, $k_\perp$, $k_\parallel$, $\delta\theta$ and $\delta k_\parallel$ is obtained and can be used as input for FDC.

4.2 Calculation of diffusion coefficients

Ni et al. (2016) argue that a new method is required to quantify pitch angle scattering of plasma sheet electrons by highly oblique, broadband electrostatic ECH emissions as opposed to chorus waves. Based on Lyons (1974), the local pitch angle diffusion coefficient due to electrostatic ECH waves is given by

$$D_{\alpha\alpha} = \sum_{n=-\infty}^{\infty} \int k_\perp \, \mathrm{d}k_\perp \left[\Psi_{n,k} \left(\frac{n\Omega_e/\omega_k - \sin^2\alpha}{\sin\alpha\cos\alpha} \right)^2 \right]_{k_\parallel = k_{\parallel,res}}, \tag{24}$$

$$\Psi_{n,k} = \frac{1}{4\pi} \frac{e^2}{m_e^2} \frac{|E_k|^2}{V} \left(\frac{\omega}{|k|} \right)^2 \frac{J_n^2(k_\perp v_\perp/\Omega_e)}{v^4 |v_\parallel - \partial\omega/\partial k_\parallel|}, \tag{25}$$

where $k_\perp$ and $k_\parallel$ are the components of the wave vector perpendicular and parallel to the ambient magnetic field B_0, respectively, $k_{\parallel,res} = (\omega_k - n\Omega_e/\gamma)/v_\parallel$ is the resonant parallel wave number, $\Omega_e = |eB_0/m_e|$ is the angular electron gyrofrequency, ω_k is the wave frequency as a function of k, γ is the Lorentz factor, V is the plasma volume and J_n is the Bessel function of order n.
Horne and Thorne (2000) developed Equation 24 into a modified version,

$$\begin{aligned} D_{\alpha\alpha} =& \frac{\sqrt{\pi}}{2} \frac{e^2}{m_e^2} \frac{|E_\omega|^2 \sin^2\varphi}{k_{0,\perp}^2 \delta l_\parallel} \frac{\exp(-\mu)}{v^5 \cos\alpha} \\ & \cdot \sum_{N=-\infty}^{+\infty} \left(\frac{n\Omega_e/\omega_k - \sin^2\alpha}{\sin\alpha\cos\alpha} \right)^2 I_n(\mu) \left\{ \exp\left[-(\zeta_n^-)^2 \right] + \exp\left[-(\zeta_n^+)^2 \right] \right\}, \end{aligned} \tag{26}$$

where $\delta k_\parallel$ is the width of the wave spectrum distribution over parallel wave number and $I_n(\mu)$ is the modified Bessel function with the argument $\mu = k_{0,\perp} v_\perp^2/(2\Omega_e^2)$ and $\zeta_n^\pm = \frac{\omega_k - n\Omega_e}{\delta k_\parallel v \cos\alpha} \pm \frac{k_{0,\parallel}}{\delta k_\parallel}$.
This can be done assuming that the parallel group velocity is small compared to the

electron parallel velocity (i.e. $\partial\omega_k/\partial k_\parallel \ll v_\parallel$) and that the electric field spectrum has the form of

$$|E_k|^2 = C' k_\perp^2 \exp\left[-\left(\frac{k_\perp}{k_{0,\perp}}\right)^2\right] \cdot \left\{\exp\left[-\left(\frac{k_\parallel - k_{0,\parallel}}{\delta k_\parallel}\right)^2\right] + \exp\left[-\left(\frac{k_\parallel + k_{0,\parallel}}{\delta k_\parallel}\right)^2\right]\right\} \tag{27}$$

with a normalization constant

$$C = \frac{4\pi^{3/2}}{k_{0,\perp}^4 \delta k_\parallel} V |E_\omega|^2 \tag{28}$$

obtained from $\int |E_\omega|^2 \, \mathrm{d}r = \frac{1}{8\pi^3} \int |E_k|^2 \, \mathrm{d}k$.
The local pitch angle-momentum mixed diffusion rate $D_{\alpha p}$ and momentum diffusion rate D_{pp} can then be obtained from $D_{\alpha\alpha}$ using Equation 22. Quasi-linear bounce-averaged diffusion coefficients can subsequently be calculated with Equations 12-14.
The workflow of the modified version of the Full Diffusion Code for ECH wave-induced scattering rates is sketched in Figure 2. Given that the wave electric field and wave normal angle distribution are known, the bounce-averaged resonant diffusion coefficients can be evaluated for ECH waves at any specified frequency. Theoretically, the quantification of diffusion rates requires an integration over the entire ECH frequency band, which depends on solving the hot plasma dispersion relation. Ni et al. (2011b) have developed an approximate method instead, using the observed ECH wave power spectrum to introduce weighting factors for the diffusion rates at each wave frequency. For each ECH harmonic band, the overall bounce-averaged diffusion rates are computed by

$$\langle D \rangle_{\text{overall}} = \sum_{j=1}^{M} R_j \langle D \rangle_j \tag{29}$$

with the weighting factor for the jth wave frequency given by

$$R_j = \frac{(I_E)_j}{\sum_{j=1}^{M} (I_E)_j}. \tag{30}$$

Here, M is the number of the frequency considered in each band, $\langle D \rangle_j$ is the bounce-averaged diffusion rate due to the jth wave frequency and $(I_E)_j$ is the electric field intensity for the jth wave frequency, given by

$$I_E(f) = A \exp\left[-\left(\frac{f - f_m}{\Delta f}\right)^2\right], \quad (f_{lc} < f < f_{uc}) \tag{31}$$

when assuming a Gaussian frequency distribution, with f_m and Δf being the frequency of maximum wave power and bandwidth, respectively, and f_{lc} and f_{uc} being the lower and upper cutoffs to the wave spectrum outside which the wave power is assumed to be zero. A is a normalization factor given by

$$A = \frac{E_\omega^2}{\Delta f} \frac{1}{\pi^{3/2}} \left[\mathrm{erf}\left(\frac{f_m - f_{lc}}{\Delta f}\right) + \mathrm{erf}\left(\frac{f_{uc} - f_m}{\Delta f}\right)\right]^{-1}, \tag{32}$$

where E_ω is the wave electric field amplitude.

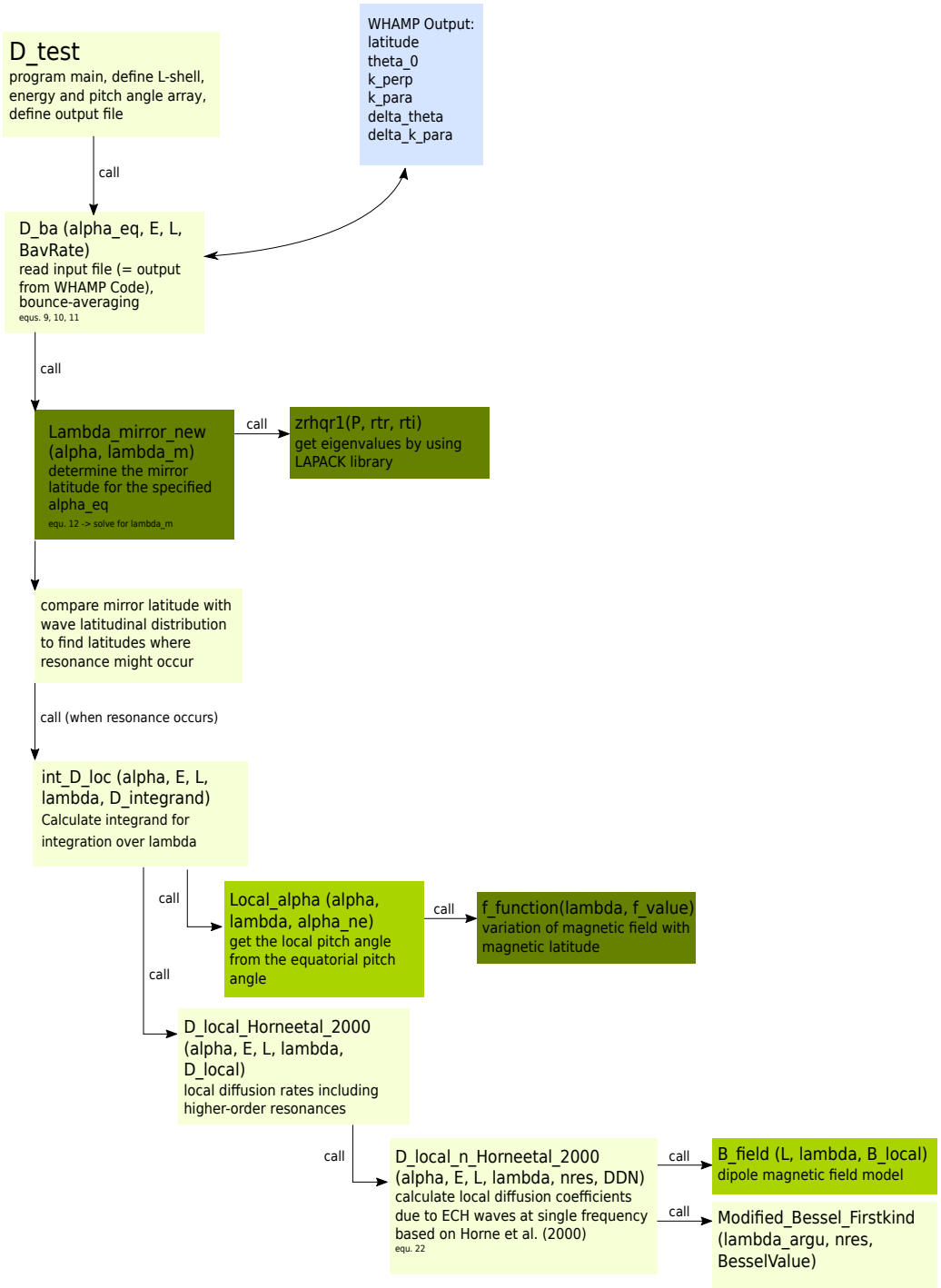


Figure 2: Full Diffusion Code for the calculation of quasi-linear bounce-averaged diffusion coefficients for wave-particle interaction with ECH waves.

5 Sensitivity of ECH wave scattering rates to hot plasma components

In general, the dispersion relation, excitation and amplification of the ECH waves are dependent on the distribution profiles of hot and cold electrons. The growth rates and wave propagation characteristics modeled by Ni et al. (2011b) and described in Section 4.1 are therefore not applicable to all conditions.
In the following, different case studies are summarized, which indicate parameters of the hot electron components, either obtained by fitting electron distributions observed by various spacecrafts at different locations by Equation 23, or from assumptions about the plasma sheet.
Horne and Thorne (2000) model the distribution with two electron components, one hot and one cold, to represent the conditions in the plasma sheet at approximately $L = 7$. Only the hot component includes a loss cone and the parameters are listed in Table 2.

Table 2: Electron components used by Horne and Thorne (2000) to model the electron distribution function at $L = 7$.

Component	$T_\perp$ / eV	$T_\parallel$ / eV	N_e / m^{-3}	Δ	β
1 (cold)	1	1	1.0×10^6	1	0
2 (hot)	1000	1000	5×10^5	0.5	0.02

Lou et al. (2022) adopt the set of parameters established by Horne and Thorne (2000), but assume a total electron density of $N_t = 4.96\,\mathrm{cm}^{-3}$ derived from the plasma trough density model in Sheeley et al. (2001). The density ratio between hot and cold electrons is set to be 1/2 and they assume the wave growth to be centered at the wave normal angle $\theta = 89.5\,^\circ$ with an angular width of $\Delta\theta = 0.5\,^\circ$. The parameters used to model the distribution function are given in Table 3. Lou et al. (2022) test the sensitivity of the diffusion coefficients to the background magnetic field, the total plasma density and the density ratio between the hot and cold electron component. They calculate pitch angle and momentum diffusion coefficients for the first 3 harmonic frequency bands and use the frequency of the peak growth rate in each band as an input to FDC. Their results show that the variation of the parameters can significantly change the magnitude of the diffusion coefficients of electrons with energies $< 1\,\mathrm{keV}$. An increase of the magnetic field intensity or the density ratio weakens the scattering efficiency and narrows the resonant region to lower pitch angles, while an increase of the total electron density has the opposite effect in this electron energy range.

Table 3: Electron components used by Lou et al. (2022) to model the electron distribution function at $L = 6$.

Component	$T_\perp$ / eV	$T_\parallel$ / eV	N_e / m^{-3}	Δ	β
1 (cold)	1	1	3.31×10^6	1	0.5
2 (hot)	1000	1000	1.65×10^6	0.5	0.02

Ni et al. (2012) evaluate ECH wave driven resonant scattering waves in the outer magnetosphere at $L = 11.5$. They fitted the electron phase space density distribution observed from the THEMIS spacecraft by the distribution given in Equation 23 and obtain the parameters listed in Table 4. They follow the method described in Section 4.1 to obtain the latitudinal variations of the wave parameters and calculate bounce-averaged diffusion rates as described in Section 4.2.

Table 4: Hot electron components used by Ni et al. (2012) to model the electron distribution function at $L = 11.5$.

Component	$T_\perp$ / eV	$T_\parallel$ / eV	N_e / m^{-3}	Δ	β
1	2.0	2.3	8.0×10^5	1.0	0.5
2	10.2	106.4	3.1×10^4	0.8	0.1
3	160.1	569.6	1.0×10^5	0.01	0.01
4	1422.1	3441.0	1.2×10^5	0.3	0.1
5	3044.5	4432.1	9.2×10^4	0.1	0.9

Fukizawa et al. (2020) determine electron temperature ($T_\perp$, $T_\parallel$), electron density N_e, loss cone depth Δ and loss cone width β by fitting the phase space density obtained by the Arase satellite. The fitting parameters are shown in Table 5. By varying the WNA from 85.0 ° to 89.5 °, they solve the hot plasma dispersion relation with the Kyoto University Plasma Dispersion Analysis Package (KUPDAP) and subsequently calculate bounce-averaged pitch-angle diffusion coefficients near the loss cone.

Table 5: Electron components used by Fukizawa et al. (2020) to model the electron distribution function at $L = 6.1$.

Component	$T_\perp$ / eV	$T_\parallel$ / eV	N_e / m^{-3}	Δ	β
1 (cold)	1-10	1-10	1.9×10^6	1	0.0
2	130	57	0.18×10^6	0.9	0.015
3	630	440	0.16×10^6	0.82	0.019
4	3.1×10^3	350	0.12×10^6	0.73	0.010
5	3.7×10^3	4.2×10^3	0.072×10^6	0.63	2.0×10^{-3}
6	1.8×10^4	5.0×10^3	0.33×10^6	0.20	0.016

5.1 Hot plasma temperature, $T_\perp = T_\parallel$

We calculate diffusion coefficients for different hot electron component models in order to test the sensitivity of the scattering rates to the different parameters. We start by taking the model from Lou et al. (2022) (Table 3) with one hot and one cold component and vary the temperature of the hot electron component. As $T_\perp$ and $T_\parallel$ of the hot components seem to be roughly in the range between 100 eV and 10 keV in all the models listed above, we take these values as boundaries. We assume no temperature anisotropy ($T_\perp = T_\parallel$) and use 20 different values ranging from 100 eV to 10 keV as input for WHAMP. We vary the wave normal angle between 86 ° and 89.96 ° and adopt the model for the

WNA distribution from Ni et al. (2011b). The parameters used to model the electron distribution are summarized in Table 6.

Table 6: Electron components used to model the electron distribution function and test the sensitivity of the diffusion coefficients to the hot plasma temperature.

Component	$T_\perp$ / eV	$T_\parallel$ / eV	N_e / m^{-3}	Δ	β
1 (cold)	1	1	3.31×10^6	1	0.5
2 (hot)	100 - 10000	100 - 10000	1.65×10^6	0.5	0.02

Using the wave parameters obtained from solving the hot plasma dispersion relation and implementing the WNA distribution, we calculate bounce-averaged quasi-linear diffusion coefficients at nine individual frequencies for each model of the hot electron distribution. An example for the pitch-angle diffusion coefficients $\langle D_{\alpha\alpha} \rangle$ for $T_\perp = T_\parallel = 1142\,\mathrm{eV}$ is shown in Figure 3. It shows that ECH waves can efficiently scatter electrons in the energy range from $< 100\,\mathrm{eV}$ to several keV over a broad range of pitch angles from $0\,^\circ$ to $\sim 60\,^\circ$, varying with frequency. The most efficient scattering appears to be at energies of several hundred eVs. At higher energies $\geq 1\,\mathrm{keV}$, the range of pitch angles for which efficient scattering can occur narrows significantly.

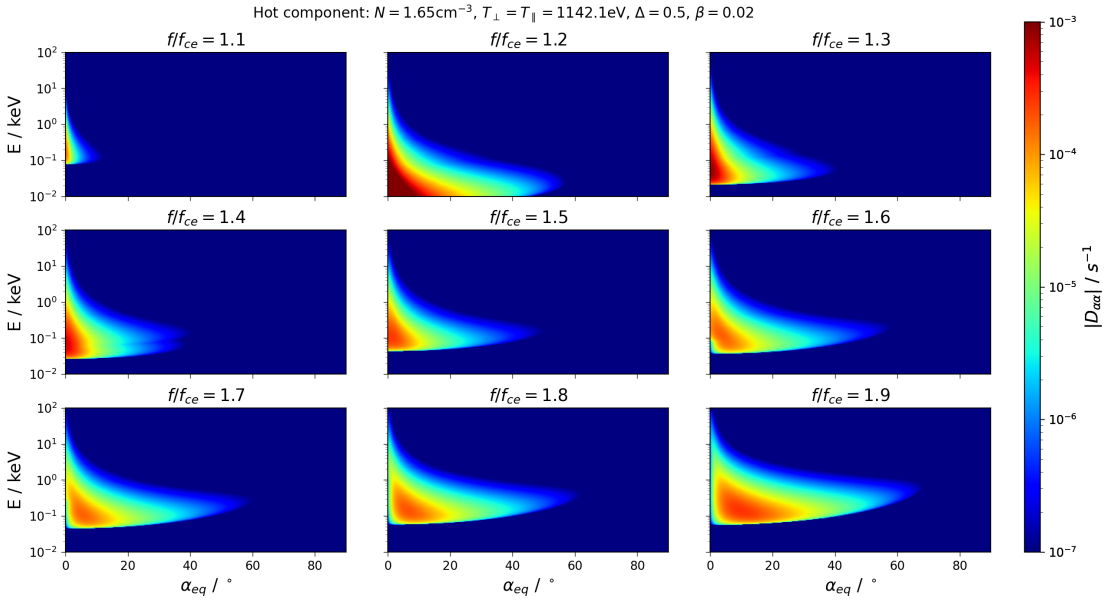


Figure 3: Quasi-linear bounce-averaged pitch angle diffusion coefficients $\langle D_{\alpha\alpha} \rangle$ for ECH wave scattering in the first harmonic band at nine individual frequencies. The electron components used to model the electron distribution function are indicated in Table 6 and $T_\perp = T_\parallel = 1142\,\mathrm{eV}$ for the hot component.

When increasing the temperature, the magnitude of the diffusion coefficients $\langle D_{\alpha\alpha} \rangle$ decreases at all frequencies $f \geq 1.3 f_{\mathrm{ce}}$. Additionally, the resonant region of the electron pitch angle narrows. An example at $T_\perp = T_\parallel = 1142\,\mathrm{eV}$ is shown in Figure 4.

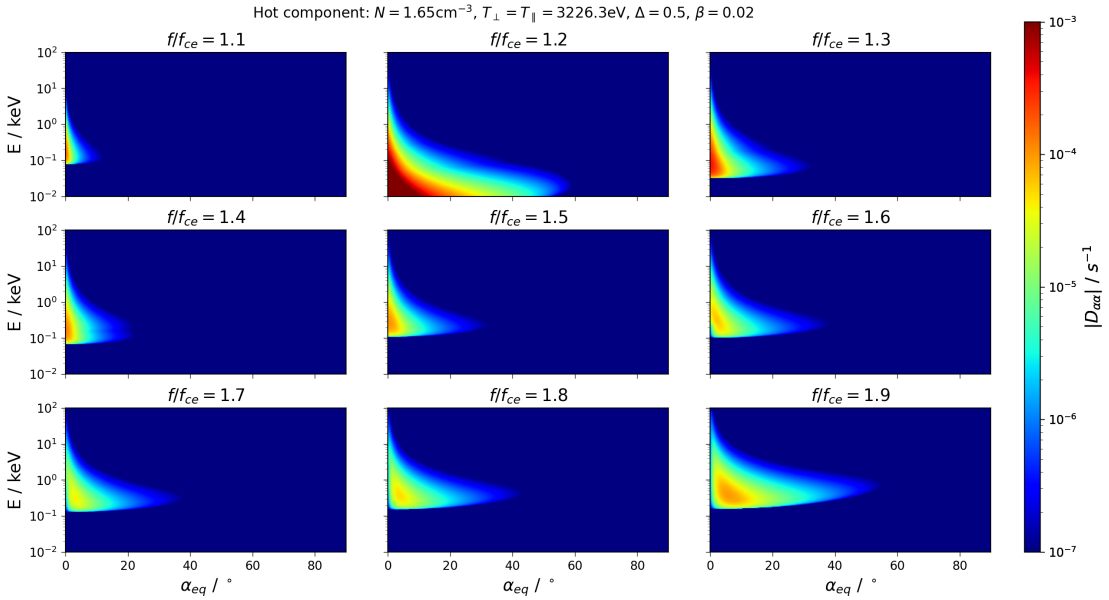


Figure 4: Quasi-linear bounce-averaged pitch angle diffusion coefficients $\langle D_{\alpha\alpha} \rangle$ for ECH wave scattering in the first harmonic band at nine individual frequencies. The electron components used to model the electron distribution function are indicated in Table 6 and $T_\perp = T_\parallel = 3226\,\mathrm{eV}$ for the hot component.

In Figures 5 to 10, the wave growth rate is shown as a function of wave normal angle for the different temperatures of the hot electron component. As described in Section 4.1, for each individual model, the wave normal angle of the peak growth rate is taken as the equatorial WNA in the model of the WNA distribution. While the growth rate peaks at high WNAs near 90 ° for all temperatures > 1000 eV at frequencies $f = 1.5\,\mathrm{f_{ce}}$ and $f = 1.8\,\mathrm{f_{ce}}$, there appears to be a second peak at smaller WNAs around 89 ° for $f = 1.2\,\mathrm{f_{ce}}$ which grows with increasing hot electron temperature. This shift of the WNA of the peak growth rate changes the WNA distribution model significantly, leading to a difference in the energy range over which efficient pitch angle scattering can occur. As shown in Figure 4, at $f = 1.2\,\mathrm{f_{ce}}$, the magnitude of the pitch angle diffusion coefficient $\langle D_{\alpha\alpha} \rangle$ is highest in the electron energy range between 10 eV and 100 eV and the range of equatorial pitch angles of the electrons which can be scattered is much larger. The shift of the WNA of the peak growth rate also occurs at other frequencies, $f = 1.1\,\mathrm{f_{ce}}$, $f = 1.3\,\mathrm{f_{ce}}$ and $f = 1.4\,\mathrm{f_{ce}}$, when increasing the temperature of the hot electron component even further. This becomes evident in Figure 11, which shows the pitch angle diffusion coefficients $\langle D_{\alpha\alpha} \rangle$ for $T_\perp = T_\parallel = 8958\,\mathrm{eV}$.
Figure 12 shows the average pitch-angle diffusion coefficients of all 20 different temperature values used to model the hot electron distribution. From this plot, we can infer the energy and pitch angle range for which electrons can be scattered efficiently in the tested temperature range.

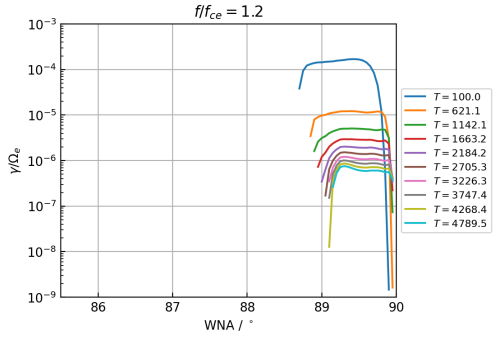


Figure 5: Growth rate as a function of wave normal angle for wave frequency $f = 1.2\,\mathrm{f_{ce}}$ and different $T_\perp = T_\parallel$.

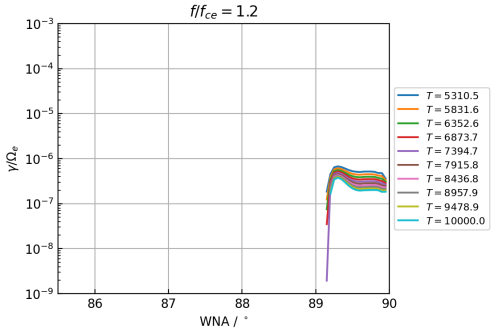


Figure 6: Growth rate as a function of wave normal angle for wave frequency $f = 1.2\,\mathrm{f_{ce}}$ and different $T_\perp = T_\parallel$.

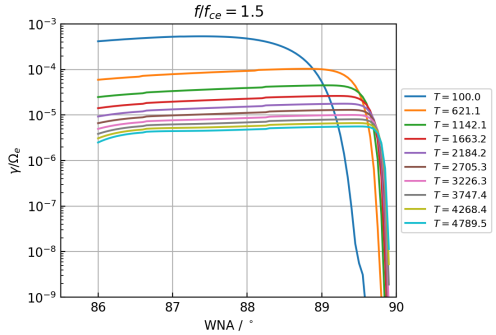


Figure 7: Growth rate as a function of wave normal angle for wave frequency $f = 1.5\,\mathrm{f_{ce}}$ and different $T_\perp = T_\parallel$.

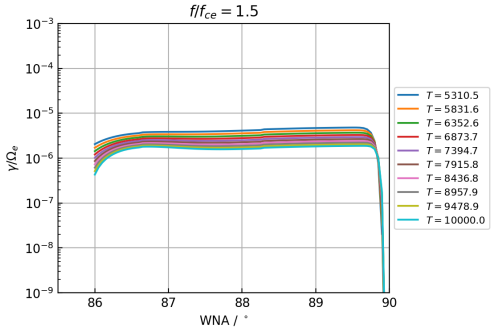


Figure 8: Growth rate as a function of wave normal angle for wave frequency $f = 1.5\,\mathrm{f_{ce}}$ and different $T_\perp = T_\parallel$.

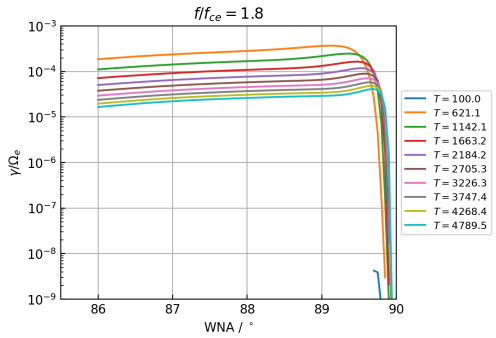


Figure 9: Growth rate as a function of wave normal angle for wave frequency $f = 1.8\,\mathrm{f_{ce}}$ and different $T_\perp = T_\parallel$.

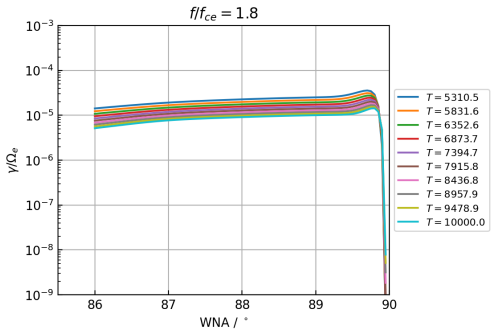


Figure 10: Growth rate as a function of wave normal angle for wave frequency $f = 1.8\,\mathrm{f_{ce}}$ and different $T_\perp = T_\parallel$.

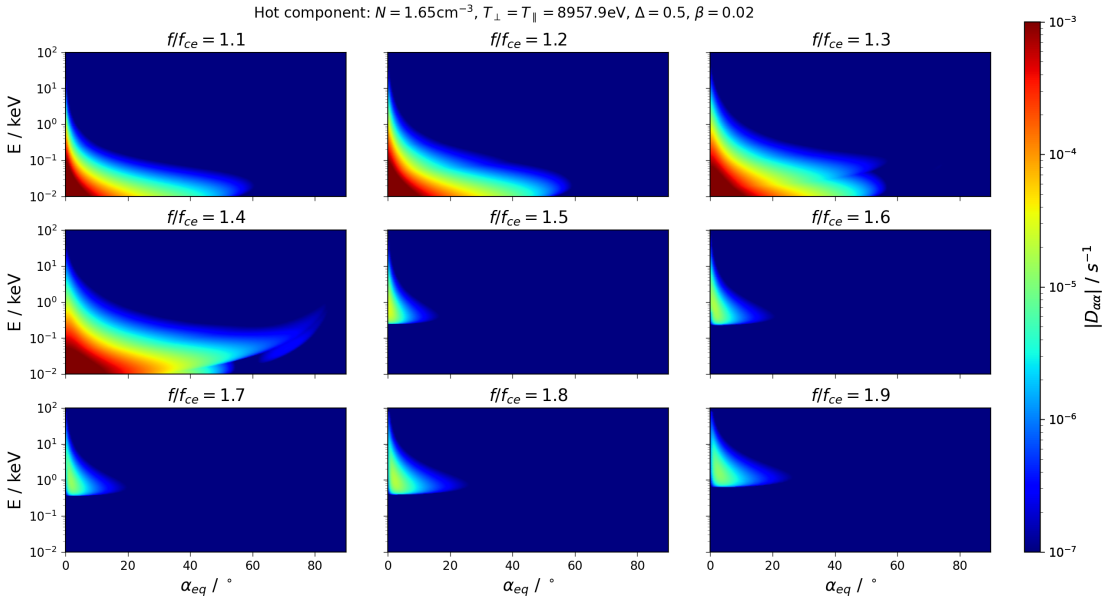


Figure 11: Quasi-linear bounce-averaged pitch angle diffusion coefficients $\langle D_{\alpha\alpha} \rangle$ for ECH wave scattering in the first harmonic band at nine individual frequencies. The electron components used to model the electron distribution function are indicated in Table 6 and $T_\perp = T_\parallel = 8436\,\mathrm{eV}$ for the hot component.

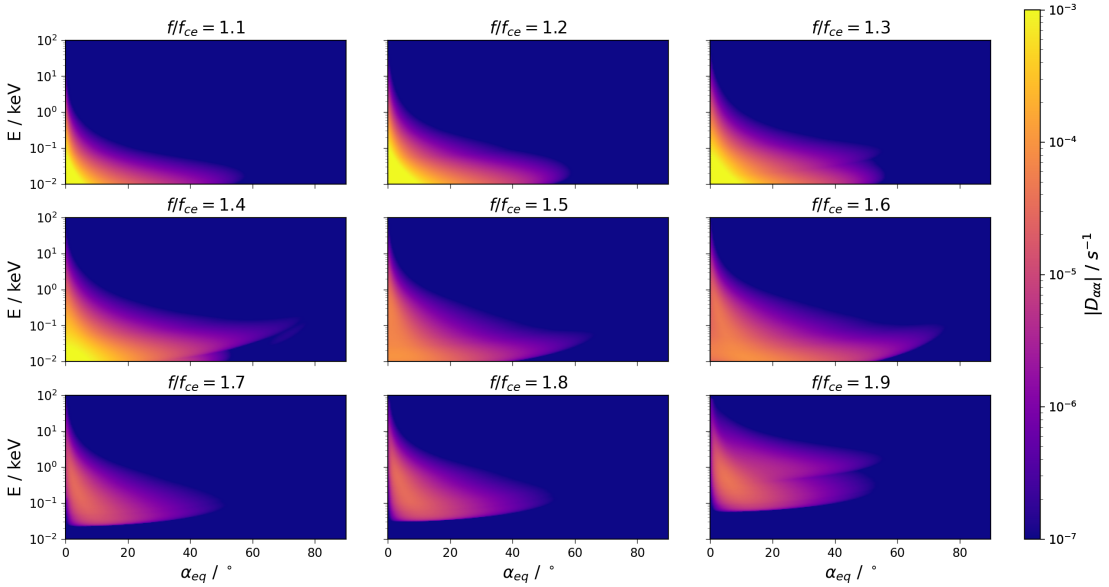


Figure 12: Quasi-linear bounce-averaged pitch angle diffusion coefficients $\langle D_{\alpha\alpha} \rangle$ for ECH wave scattering in the first harmonic band at nine individual frequencies. The electron components used to model the electron distribution function are indicated in Table 6. This plot shows the average of the diffusion coefficients modeled with all 20 different hot electron temperatures.

5.2 Hot plasma temperature, with temperature anisotropy

Next, we calculate diffusion coefficients for a hot electron distribution with a temperature anisotropy. In this case, $T_\perp$ is varied from $T_\perp = 100\,\mathrm{eV}$ to $T_\perp = 10\,\mathrm{keV}$, while $T_\parallel$ goes from $T_\parallel = 10\,\mathrm{keV}$ to $T_\parallel = 100\,\mathrm{eV}$ at the same time. Again, we model 20 different temperature steps and use the model from Lou et al. (2022) for all other parameters. An example for the pitch-angle diffusion coefficients $\langle D_{\alpha\alpha} \rangle$ for $T_\perp = 2184\,\mathrm{eV}$ and $T_\parallel = 7916\,\mathrm{eV}$ is shown in Figure 13. Again, electrons at energies of several hundred eVs seem to be scattered most efficiently. Compared to Figure 3, the range of pitch angles where scattering occurs is much smaller, extending from $0\,^\circ$ to $\sim 20\,^\circ$.

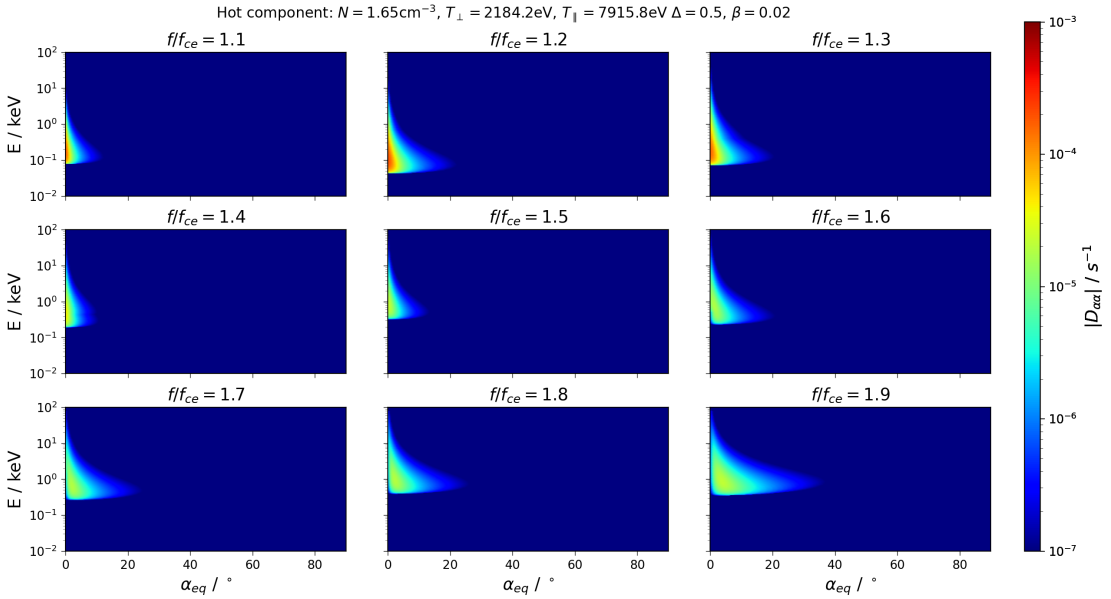


Figure 13: Quasi-linear bounce-averaged pitch angle diffusion coefficients $\langle D_{\alpha\alpha} \rangle$ for ECH wave scattering in the first harmonic band at nine individual frequencies. The electron components used to model the electron distribution function are indicated in Table 6 and $T_\perp = 2184\,\mathrm{eV}$ and $T_\parallel = 7916\,\mathrm{eV}$ for the hot component.

With increasing $T_\perp$ and decreasing $T_\parallel$, there is again a shift of the wave normal angle of the peak growth rate for $f = 1.2\,\mathrm{f_{ce}}$. The wave growth rate as a function of wave normal angle for the different temperature anisotropies is shown in Figures 14 to 19 for the frequencies $f = 1.2\,\mathrm{f_{ce}}$, $f = 1.5\,\mathrm{f_{ce}}$ and $f = 1.8\,\mathrm{f_{ce}}$. In Figure 14, it becomes clear that the growth rate peaks at a high wave normal angle close to $90\,^\circ$ for $T_\perp/T_\parallel = 0.01$, and that this peak shifts to smaller WNAs close to $89\,^\circ$ when $T_\perp/T_\parallel$ increases. This shift leads to a different wave normal angle model, which changes the resonant energy and pitch angle range. The diffusion coefficients $\langle D_{\alpha\alpha} \rangle$ for $T_\perp = 3226\,\mathrm{eV}$ and $T_\parallel = 6874\,\mathrm{eV}$ are shown in Figure 20. Comparing the Figures 13 and 20, the change of the resonant scattering range at $f = 1.2\,\mathrm{f_{ce}}$ becomes evident.

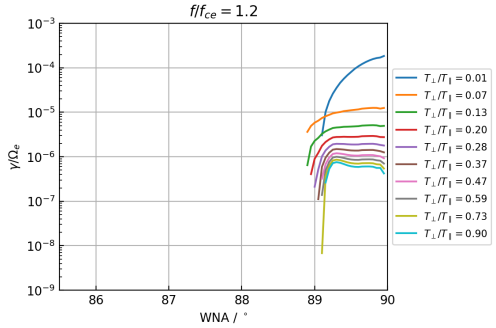


Figure 14: Growth rate as a function of wave normal angle for wave frequency $f = 1.2\,\mathrm{f_{ce}}$ and different $T_\perp/T_\parallel$.

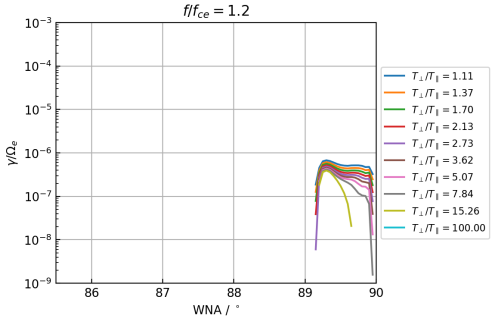


Figure 15: Growth rate as a function of wave normal angle for wave frequency $f = 1.2\,\mathrm{f_{ce}}$ and different $T_\perp/T_\parallel$.

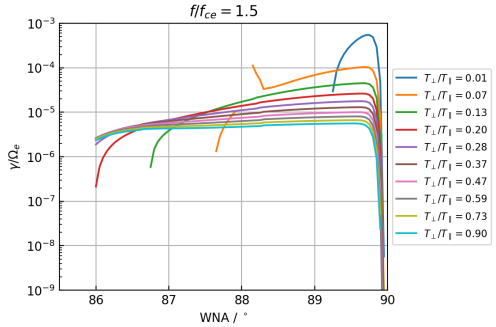


Figure 16: Growth rate as a function of wave normal angle for wave frequency $f = 1.5\,\mathrm{f_{ce}}$ and different $T_\perp = T_\parallel$.

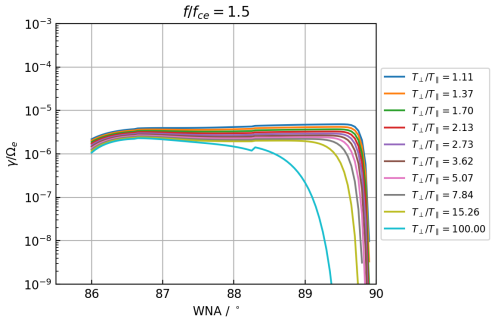


Figure 17: Growth rate as a function of wave normal angle for wave frequency $f = 1.5\,\mathrm{f_{ce}}$ and different $T_\perp = T_\parallel$.

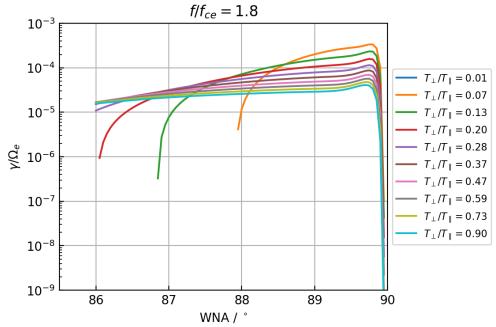


Figure 18: Growth rate as a function of wave normal angle for wave frequency $f = 1.8\,\mathrm{f_{ce}}$ and different $T_\perp = T_\parallel$.

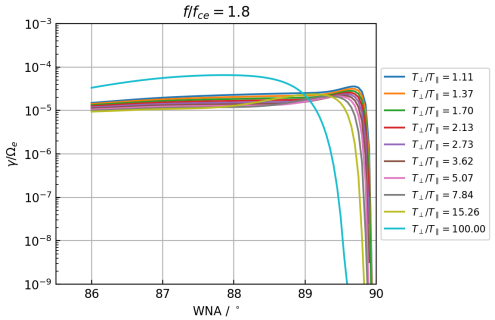


Figure 19: Growth rate as a function of wave normal angle for wave frequency $f = 1.8\,\mathrm{f_{ce}}$ and different $T_\perp = T_\parallel$.

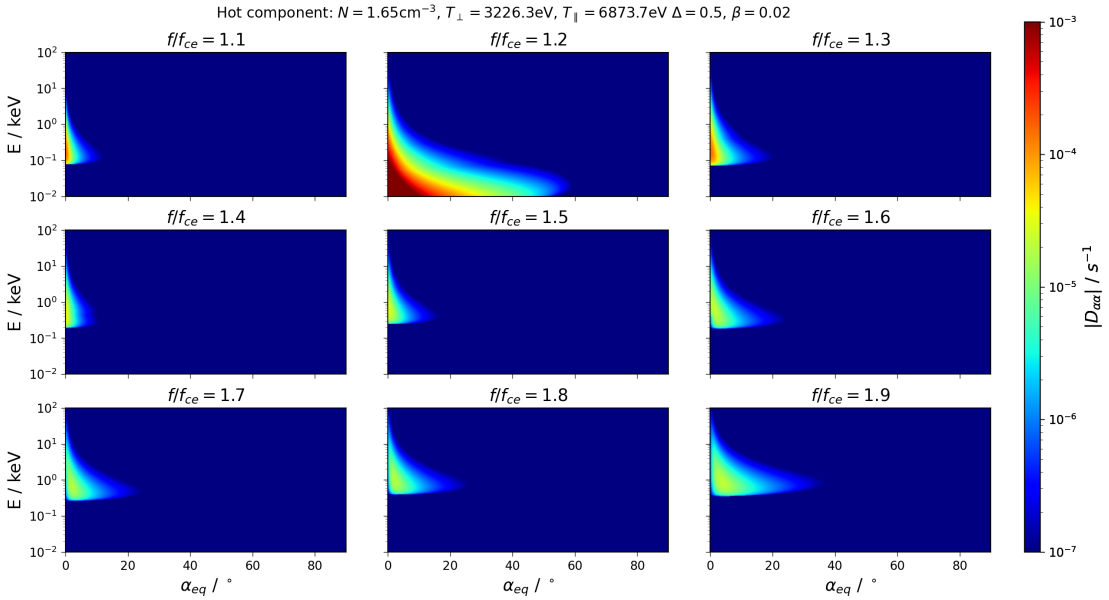


Figure 20: Quasi-linear bounce-averaged pitch angle diffusion coefficients $\langle D_{\alpha\alpha} \rangle$ for ECH wave scattering in the first harmonic band at nine individual frequencies. The electron components used to model the electron distribution function are indicated in Table 6 and $T_\perp = 3226\,\mathrm{eV}$ and $T_\parallel = 6874\,\mathrm{eV}$ for the hot component.

When further increasing $T_\perp/T_\parallel$, the magnitude of the diffusion coefficients increases and the scattering occurs over a broader range of pitch angles and energies. Additionally, the peak growth rate shifts to lower wave normal angles not only at $f = 1.2\,\mathrm{f_{ce}}$, but also at other frequencies. An example for $T_\perp = 9479\,\mathrm{eV}$ and $T_\parallel = 621\,\mathrm{eV}$ is shown in Figure 21. Figure 22 shows the average pitch-angle diffusion coefficients of all 20 different values for $T_\perp/T_\parallel$ used to model the hot electron distribution. Comparing Figures 12 and 22, we see that electrons with a broader range of pitch-angles and energies can be scattered at wave frequencies $f \geq 1.5\,\mathrm{f_{ce}}$ when the hot temperature component is anisotropic.

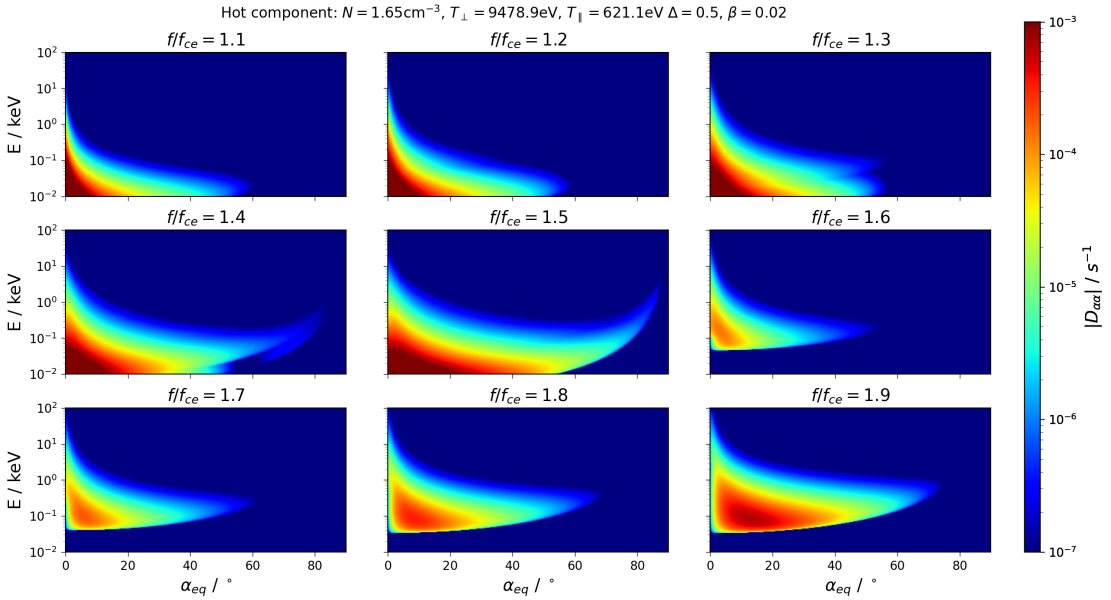


Figure 21: Quasi-linear bounce-averaged pitch angle diffusion coefficients $\langle D_{\alpha\alpha} \rangle$ for ECH wave scattering in the first harmonic band at nine individual frequencies. The electron components used to model the electron distribution function are indicated in Table 6 and $T_\perp = 9479\,\mathrm{eV}$ and $T_\parallel = 621\,\mathrm{eV}$ for the hot component.

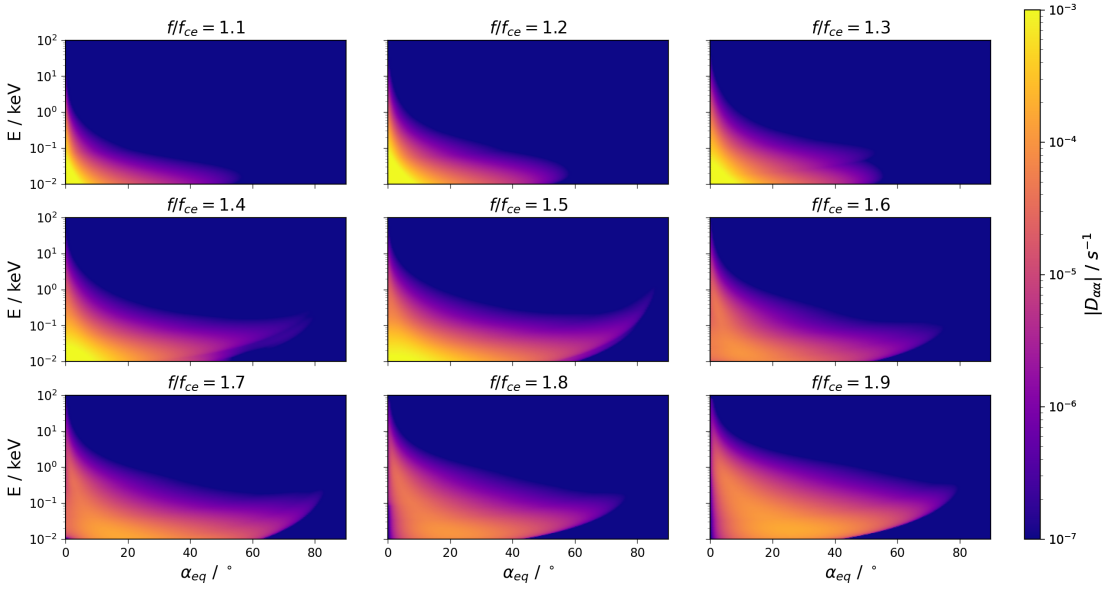


Figure 22: Quasi-linear bounce-averaged pitch angle diffusion coefficients $\langle D_{\alpha\alpha} \rangle$ for ECH wave scattering in the first harmonic band at nine individual frequencies. The electron components used to model the electron distribution function are indicated in Table 6. This plot shows the average of the diffusion coefficients modeled with all 20 different hot electron temperatures.

6 Conclusion and Outlook

Several questions remain concerning the formation processes of diffuse aurora.
While whistler-mode chorus waves have been identified as the dominant source for pitch angle scattering in the inner magnetosphere ($L < 8$), the exact role of ECH waves in driving the nightside electron diffuse aurora is still unclear, especially under different geomagnetic conditions. Although observations suggest that whistler-mode chorus waves are absent and ECH waves are relatively scarce at high L-shells, the diffuse aurora persists at high latitudes which are expected to map to L-shells outside the inner magnetosphere from $L \sim 8$ to beyond $L \sim 15$. Recent studies (Zhang et al., 2013; Zhang et al., 2015, e.g.) suggest that ECH waves, possibly with small amplitude, are still present in the outer magnetosphere ($L > 8$) and could be an important driver of the nightside electron diffuse aurora, especially during geomagnetically active times.
Quasi-linear bounce-averaged diffusion coefficients can be calculated with the Full Diffusion Code for wave-particle interaction with whistler-mode chorus waves and with a modified version of the code for ECH waves. Ni et al. (2017) carried out a statistical analysis of wave spectral intensity, peak wave frequency and wave occurrence rate with respect to L-shell, magnetic local time, magnetic latitude and geomagnetic activity level based on the high-resolution FFF wave spectral data obtained from the Time History of Events and Macroscale Interactions during Substorms spacecraft. Their study provides a statistical model of ECH wave global distribution in the Earth's inner and outer magnetosphere, which can be adopted as input in the Full Diffusion Code to evaluate the rates of ECH wave-driven pitch angle scattering.
However, the diffusion coefficients additionally depend on local plasma conditions, such as the background magnetic field, plasma density, and properties of the hot plasma sheet electrons responsible for the wave excitation. Lou et al. (2022) performed a sensitivity study of the ECH wave-induced diffusion coefficients to the magnetic field intensity, total electron density and the ratio of hot to cold electron density. Their results show that the pitch-angle diffusion coefficients $\langle D_{\alpha\alpha} \rangle$ at electron energies $\leq 1\,\mathrm{keV}$ are strongly influenced by the variation of the background parameters. The scattering efficiency at higher energies is almost unaffected.
In Section 5, we calculated pitch angle diffusion coefficients for different temperatures of the hot electron component. Our results show that the hot electron temperature can change the magnitude of the diffusion coefficients over a large range of pitch angle and energy. With increasing $T_\perp = T_\parallel$, the magnitude generally decreases and the pitch angle and energy range in which efficient scattering can occur becomes smaller. Varying the temperature anisotropy, we find that with increasing $T_\perp/T_\parallel$, the magnitude and pitch angle and energy range of the scattering increase. Additionally, the hot electron temperature changes the distribution of the peak growth rate versus wave normal angle. Because the wave normal angle of the peak growth rate is taken to be the equatorial wave normal angle, which then increases with latitude, a shift of the peak growth rate changes the diffusion coefficients significantly.
In our calculations, we did not assume a wave power spectrum, but calculated pitch angle diffusion coefficients at 9 individual frequencies in the first band of the ECH waves.

Previous studies (Zhou et al., 2017) report that the spectrum of ECH waves is not necessarily centered in the middle of the frequency band, but can peak both in the lower half and in the upper half of the harmonic bands at a similar spatial location. This is because the peak frequencies of the wave growth rate increase with increasing hot electron temperature. Lou et al. (2022) assumed the wave frequency with the maximum growth rate to dominate the scattering process. However, in their calculations, the peak wave frequency approaches the higher gyroharmonic frequency, which they argue to be not quite consistent with observations.
Furthermore, we used the plasma parameter model from Lou et al. (2022), which is based on the model from Horne and Thorne (2000) and claimed to be a typical set of parameters. However, the model is not directly based on spacecraft observations. It could therefore be an improvement to our study to use a model based on fitting the observed electron distribution, such as Ni et al. (2011b), Ni et al. (2012) or Fukizawa et al. (2020).

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